MAHADEVAN KALLAT

B.Tech in Computer Science & Engineering

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PROFESSIONAL SUMMARY

AI/ML-focused Computer Science undergraduate who ships full-stack, cross-platform software end to end — from architecture to packaged release. Combines strong fundamentals in Java with hands-on experience integrating LLM APIs (Gemini, Ollama, vLLM) into production-style developer tools, using Git, Docker, and CI pipelines to take projects from prototype to shipped product.

EDUCATION

SRM Institute of Science and Technology *Chennai, India*

Bachelor of Technology in Computer Science and Engineering 2025 – 2029

ACHIEVEMENTS

- 3rd Place, Code Merge Hackathon v2.0 – AI for Impact, organized by Dept. of DSBS, SRM IST (Sept 2026) — for INTERLOCK, a deterministic security gateway for autonomous LLM agents.
- 2nd Place, INGENIA 1.0, organized by Directorate of Career Centre & Dept. of DSBS, SRM IST (Feb 2025) — as part of Team COSMOS.

TECHNICAL SKILLS

- Languages: C, Java | TypeScript, JavaScript, Rust | Python
- Desktop & Frontend: Tauri, React, Tailwind CSS, Component Architectures
- AI & LLM Tools: Google Gemini API, Ollama (Local Inference), vLLM, Prompt Engineering
- Computer Vision & ML: OpenCV, MediaPipe, scikit-learn, Real-Time Inference Pipelines
- DevOps & Tooling: Git, GitHub (Branching, PRs, Actions), Docker, VS Code, NSIS Packaging
- Operating Systems: Linux (Manjaro, Bazzite, Fedora/Arch environments), Windows 10/11

FEATURED PROJECTS

GestureForge | Real-Time Gesture Recognition Suite *Python, OpenCV, MediaPipe, scikit-learn*

- Built a native real-time dual-hand gesture recognition app using MediaPipe 21-keypoint tracking and a hybrid RandomForest + geometric-fallback classifier, sustaining 29+ FPS at 34ms end-to-end latency.
- Engineered 8 scale/translation-invariant geometric features (normalized distances and inter-finger angles) that lifted cross-session generalization accuracy from 80.67% to 99.67% versus raw 63-dim landmark coordinates.
- Designed a threaded, non-blocking camera pipeline and automated evidence-recording system that auto-generates structured session reports (Markdown, CSV, JSON) with FPS, latency, and confidence telemetry.

CodeLens AI | AI-Powered Desktop Code Intelligence Platform *Tauri, React, TypeScript, Ollama*

- Architected and shipped a lightweight, native desktop IDE companion app using Tauri and React, eliminating 60%+ webengine memory overhead versus an Electron-based equivalent.
- Engineered local and remote LLM integration pipelines using Ollama and vLLM for real-time contextual code reviews, bug detection, and syntax explanations.
- Implemented multi-branch Git development workflows, authored production Windows NSIS installer packages, and configured continuous release delivery.

RazorRC | Intelligent Merchant Advisory & Analytics Engine *Gemini API, React, Cross-Platform CI*

- Developed an AI advisory tool for Razorpay merchants to analyze transaction anomalies and deliver actionable reconciliation recommendations via structured prompts.

- Integrated the Gemini API with token streaming and failover handling to provide high-speed automated business intelligence summaries.
- Configured cross-platform build pipelines for Windows, Linux, and macOS with end-to-end user documentation and release lifecycle management.

PROJECT EXPERIENCE & OPEN-SOURCE COLLABORATION

Independent Software Engineering & Systems Development

2026 – Present

- Proficient in prompt engineering and LLM-assisted development for rapidly building and refactoring codebases across languages (Rust, TypeScript, Python) and frameworks.
- Practiced industry-standard Git collaboration, including atomic pull requests, peer code reviews, branching models, and detailed issue tracking across personal repositories.
- Demonstrated agile problem-solving by independently debugging Linux kernel-level environment setups, cross-compilation toolchains, and multi-architecture build scripts.

KEY HIGHLIGHTS & COMPETENCIES

- Cross-Platform Delivery: Packaged native desktop applications across Windows and Linux distributions.
- Linux Systems Proficiency: Daily-driver experience configuring advanced Linux distributions (Manjaro, Bazzite).
- Rapid Tech Adoption: Shipped complex Rust/Tauri and Python/CV projects within weeks by applying AI-assisted generation alongside high-level architectural thinking.

LANGUAGES

English: Professional • Hindi: Fluent • Malayalam: Native • Tamil: Conversational